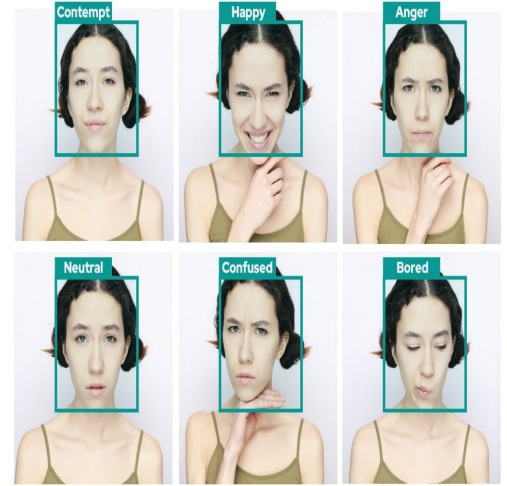


Emotion Detector

Sansshita Baskaran, Garrett Delaney, Evan Sage

Problem Statement

- Emotion recognition models are powerful but often rely on remote cloud servers to process video data
- Leads to latency, high bandwidth use, and privacy concerns
- Cloud dependence makes systems vulnerable to connectivity issues and limits real-world deployment

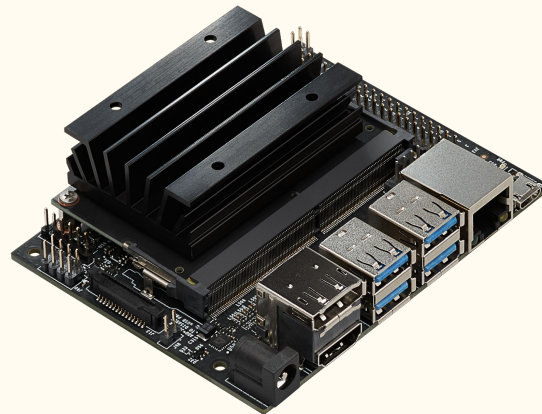


Goal and Platform

- Develop a real-time emotion detection system that works entirely on device. No internet required.
- Ensure it is fast, private, and also reliable in real-world environment.
- Platform: We chose the NVIDIA Jetson (Edge AI)

Hardware

- NVIDIA Jetson
- Zoom Q4 USB Webcam
- USB Keyboard
- USB Mouse
- HDMI Monitor



Libraries and Software Stack

- OpenCV
- TensorFlow
- Jetson Software Development Kit
- V4L2- Video for Linux



Dataset

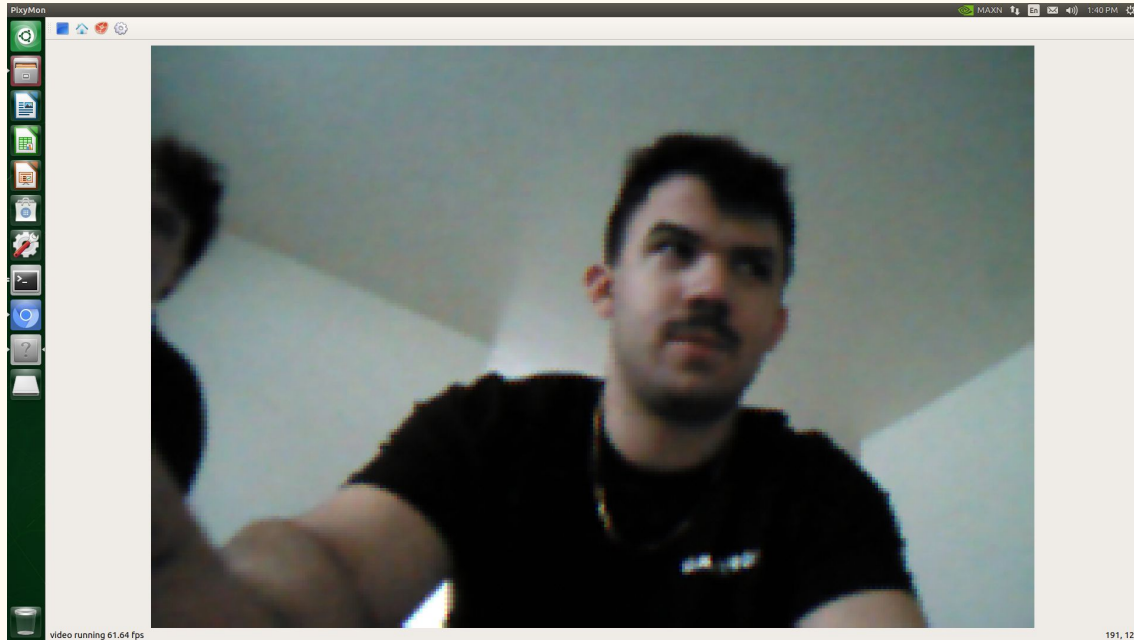


- We used FER-2013 dataset
- 48x48 pixel grayscale images of faces
- 7 Emotions: Angry, Disgusted, Fear, Happy, Neutral, Sad, Surprised
- Training set includes 28,709 examples
- Public test set includes 3,589 examples

Achievements

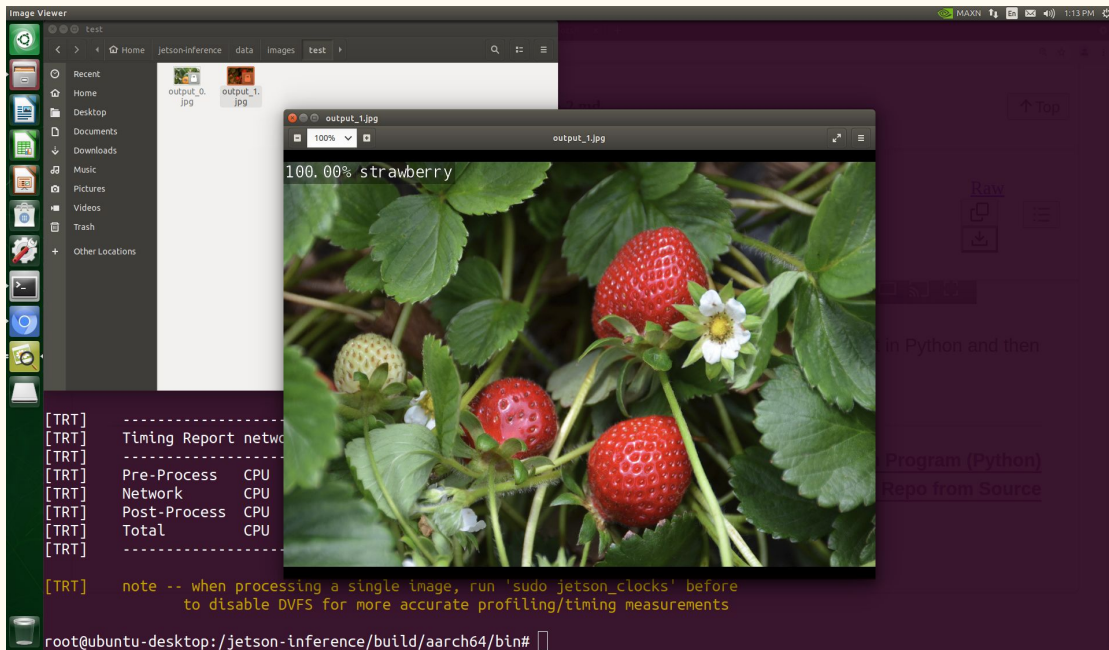
- Jetson setup complete
 - Solved Power Issues
 - Installed NVIDIA JetPack SDK
 - Verified system stability and GPU acceleration
- AI Inference Functional
 - Installed and ran Pytorch, ImageNet successfully
 - The jetson classifies test images
 - Jetson can run multiple different models for comparison
- Webcam integration working
 - Connected and configured USB webcam for live video streaming and emotion detection

Initial Progress Pictures



- Webcam is connected and streaming live video directly to the Jetson.
- Demonstrates readiness for real-time emotion detection

Progress Pictures Continued



- Jetson was able to caption an image of strawberries with 100% confidence.
- Confirms Jetson-inference and GPU acceleration work locally.

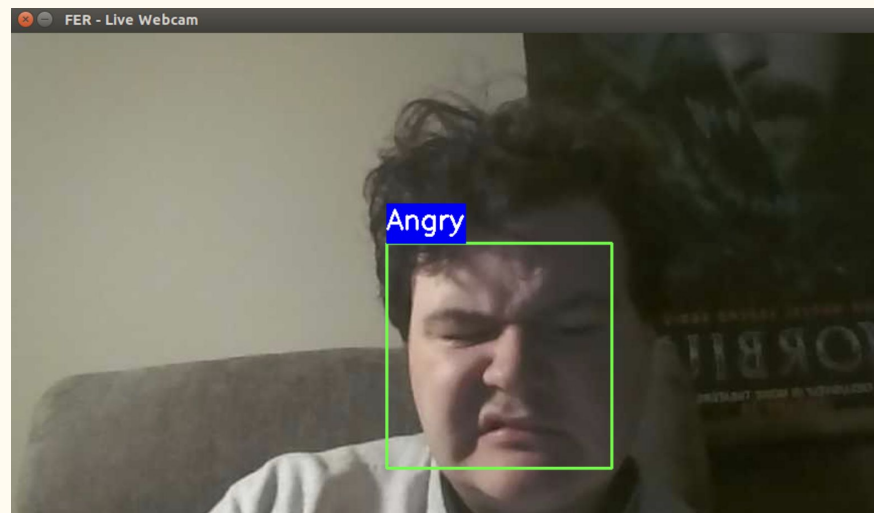
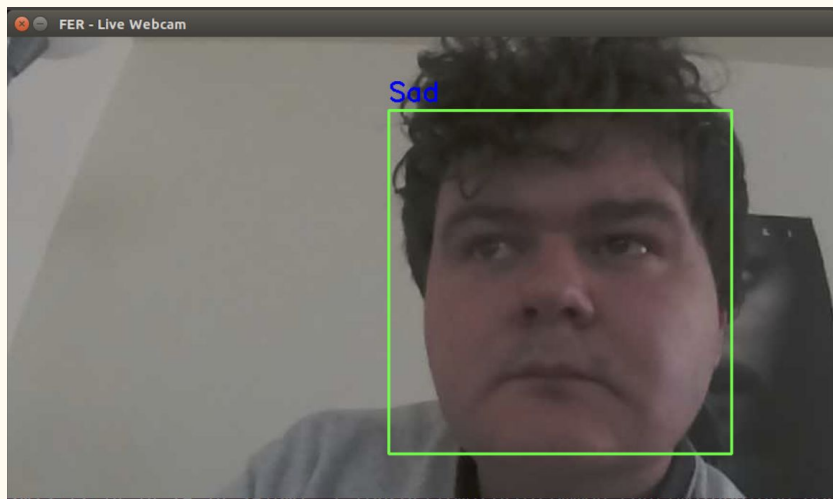
Challenges

- Original board had power issues, would not turn on
- Replaced the Jetson board with a new board
- Missed or unsupported NVIDIA libraries
- Reinstalled few libraries such as OpenCV, TensorFlow
- The board would use too much power and then shuts down
- Original Pixy Camera is not a standard USB camera, uses its own custom software, so had to rent a camera from the library
- Memory issues
 - Models were too large and not optimized for edge deployment
 - Tensorflow memory leaks, had to restart the jetson board everytime

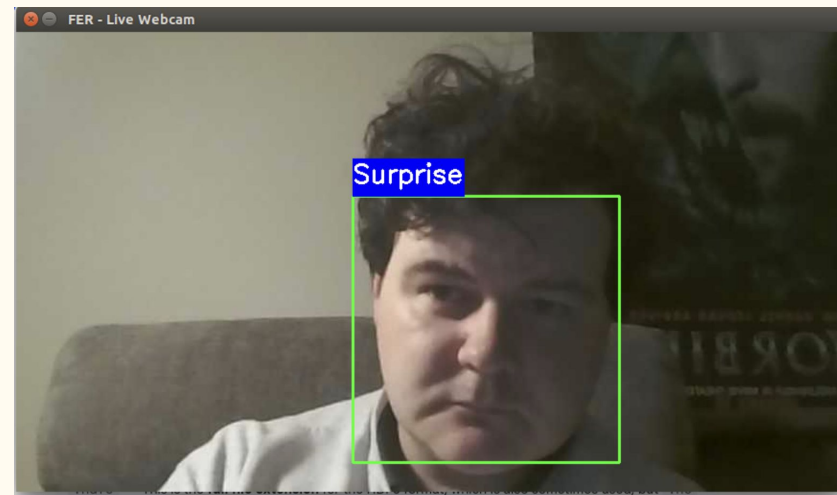
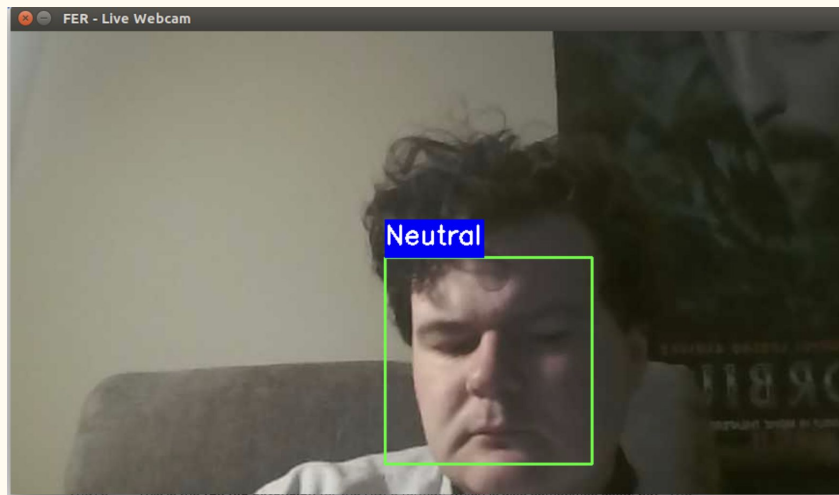
Final Results

- Worked reasonably well
- Video was extremely choppy, around 5 FPS due to memory issues
- Model favors surprise
- Certain emotions, disgust and fear, were easily overwritten by other emotions

Final Results



Final Results



Lessons Learnt

- Preprocessing is required to transition from still image to video
- Jetson platform is incredibly delicate
- Memory management and model size becomes much more important on a resource constrained device, especially one running in real time
- Nothing ever works the first time, always encounter more issues than you anticipated

Future Work

- Optimize models for a resource-constrained real time environment so that the video is less choppy (> 20 FPS)
 - e.g. Quantize the model to cut memory footprint down
- Transition to an advanced Jetson board to avoid library limitations and improve software support
- Headless setup so that it can be deployed anywhere and not need a monitor